International Biodeterioration & Biodegradation 142 (2019) 131–136

Contents lists available at ScienceDirect

International Biodeterioration & Biodegradation

journal homepage: www.elsevier.com/locate/ibiod

Long-term biogas desulfurization under different microaerobic conditions in full-scale thermophilic digesters co-digesting high-solid sewage sludge

Andrea Giordano[a], Francesco Di Capua[b,*], Giovanni Esposito[b], Francesco Pirozzi[b]

[a] Acqua & Sole s.r.l., Via Giulio Natta, Vellezzo, Bellini, PV, Italy
[b] Department of Civil, Architectural and Environmental Engineering, University of Naples Federico II, via Claudio 21, 80125, Naples, Italy

ARTICLE INFO

Keywords:
Biogas desulfurization
Microaeration
Thermophilic digestion
Full-scale digesters
Sulfide removal

ABSTRACT

Long-term (81 weeks) microaerobic biogas desulfurization was investigated in three full-scale thermophilic digesters in series co-digesting sewage sludge with co-products of source segregated domestic food waste. Residual oxygen (RO) concentrations in the digester headspaces were maintained at fixed values from 2.0% to 0.2% (v/v). Complete H_2S removal was obtained at all tested RO concentrations under organic loading rate (OLR) and hydraulic retention time (HRT) varying between 1.5 and 2.3 kg VS/m^3 d and between 44 and 71 d, respectively. Specific oxygen and air consumptions of 0.88 kg O_2/ton VS_{in} and 0.73 %ton air/ton biogas, respectively, were obtained by linear regressions with the RO values maintained in the biogas. Mass balance and stoichiometric evaluations revealed that from 13% to 59% of the consumed O_2 was used for processes other than partial H_2S oxidation to elemental sulfur (S^0). The relatively high ammonium concentrations (3.5–3.8 g N–NH_4^+/kg digestate) in the liquid phase of the three digesters presumably limited the biological H_2S production and S^0 accumulation in the headspace. The results of this study can be used for designing the aeration systems of full-scale thermophilic digesters.

1. Introduction

Anaerobic digestion (AD) is widely used for the treatment of organic waste such as agro-waste, sewage sludge and the organic fraction of municipal solid waste. During this process, almost 95% of the organic matter and 90% of the energy that originally exist in the substrate are retained as biogas (Guerrero et al., 2015). The biogas produced from the digestion of sewage sludge typically contains 55–65% of CH_4, 35–44% of CO_2 and small concentrations of other gases (Rasi et al., 2011). Hydrogen sulfide (H_2S) commonly occurs in biogas due to the anaerobic fermentation of sulfur-containing organic molecules (i.e. proteins) and the activity of sulfate-reducing bacteria (SRB) coupling the oxidation of organics and hydrogen gas (H_2) to the complete reduction of sulfate (SO_4^{2-}) and other oxidized sulfur compounds (Abatzoglou and Boivin, 2009; Papirio et al., 2013a, 2013b). The concentration of H_2S in biogas varies from a few hundred to ten thousand ppm depending on the amount of bioavailable sulfur compounds in the feedstock and the outcome of the competition among SRB, acetogens and methanogens for the organic substrates (Stams et al., 2005).

Elevated concentrations of H_2S have detrimental effects on biogas equipment and significantly reduce the economic value and reuse potential of biogas (Khanongnuch et al., 2018). H_2S conversion to sulfur oxides (SO_x) and sulfuric acid (H_2SO_4) during biogas combustion results in severe corrosion of the piping system and other industrial installations. H_2S levels in biogas should not exceed 1000 ppm if traditional boilers and internal combustion engines are utilized for electricity generation and even stricter limits (< 0.1 ppm) are required for off-site biogas utilization, e.g. as vehicle fuel or in fuel cells (Rasi et al., 2011). H_2S and SO_x are also responsible for poisoning and deactivation of the oxidation catalysts applied for biogas valorization (Mancino et al., 2016; Ortloff et al., 2016). Additionally, free H_2S levels above 50 mg/L can severely inhibit methanogenic activity (Oude Elferink et al., 1994).

Microaeration has been reported as an efficient and cost-effective method for in-situ H_2S removal from biogas (Krayzelova et al., 2014). This method consists in delivering a limited flux of air to the digester headspace in order to trigger the biological oxidation of the H_2S contained in the biogas, which is carried out by mats of sulfur-oxidizing bacteria forming on the internal walls of the headspace and on the liquid surface (Kobayashi et al., 2012). As shown by Eqs. (1) and (2) (Tilahun et al., 2018), microaerobic H_2S oxidation results in the production of SO_4^{2-} and biogenic S^0, the latter forming white deposits in the digester headspace which could be recovered and reused for water and wastewater treatment (Di Capua et al., 2019, 2016).

* Corresponding author.
E-mail address: francesco.dicapua2@unina.it (F. Di Capua).

https://doi.org/10.1016/j.ibiod.2019.05.017
Received 11 March 2019; Received in revised form 10 May 2019; Accepted 15 May 2019
Available online 21 May 2019
0964-8305/ © 2019 Elsevier Ltd. All rights reserved.

$$HS^- + 2\ O_2 \rightarrow SO_4^{2-} + H^+ \ \Delta G^\circ = -732.6\ kJ/mol \quad (1)$$

$$HS^- + 0.5\ O_2 \rightarrow S^0 + OH^- \ \Delta G^\circ = -169.4\ kJ/mol \quad (2)$$

Microaerobic conditions require a small oxygen supply, which results in limited cost and energy consumption for aeration. The potential drawbacks of microaeration include (1) methane dilution (especially if air is used instead of pure oxygen), (2) clogging of gas pipes due to S^0 accumulation and (3) inhibition of methanogenesis, although no significant oxygen inhibition on methanogenic activity was observed in previous studies (Krayzelova et al., 2015). As a result, optimization of oxygen/air dosing to the digester is necessary to maintain high quality standards for biogas production, reduce maintenance and operational costs and meet the requirements of biogas utilization technologies (Awe et al., 2017).

Full-scale anaerobic digesters require an even more accurate control of microaeration to avoid lack or surplus of oxygen in biogas due to fluctuations in feedstock composition and H_2S production. Different strategies have been proposed to control microaerobic biogas desulfurization. The oxidation reduction potential (ORP) was reported as a suitable parameter to control oxygen/air dosing when maintained at values between −310 and −230 mV (Khanal et al., 2003; Nghiem et al., 2014). In contrast, Diaz et al. (2010) observed that the relationship between ORP and micro-oxygenation was not significant and, therefore, oxygen dosing control based on ORP response was not reliable. Ramos and Fdz-Polanco (2014) proposed biogas production as a control parameter under steady sulfur loadings, whereas H_2S concentration in biogas was suggested as a suitable regulating parameter under variable sulfur loadings. However, the use of these two parameters for full-scale operation may result in technical and operational issues: (1) biogas production would be overestimated in case of comparable air and biogas flows, inducing an increase of air dosage that may disrupt the anaerobic process (Krayzelova et al., 2015); (2) a correct measurement of biogas production usually requires the installation of a biogas flow meter on long straight pipes for each digester (EN ISO 5167-2). A full-scale implementation of this configuration with multiple digesters is challenging; (3) online electrochemical sensors monitoring H_2S levels in the digester headspace commonly experience a loss of analytical sensitivity due to S^0 deposition on the electrodes (Hall and Schoenfisch, 2018; Pandey et al., 2012). As a result, air dosing control based on H_2S measurement might not be reliable for long-term operation.

Previous studies focused on the effect of microaeration on mesophilic digestion especially at pilot (Diaz et al., 2011; Ramos et al., 2014b) and full scale (Jeníček et al., 2017, 2008). In this study, the desulfurization performance of three full-scale thermophilic digesters in series was monitored for 81 weeks under different microaerobic regimes. The impact of the different microaerobic conditions on thermophilic biogas production was also evaluated.

2. Materials and methods

2.1. Full-scale plant description

The full-scale plant is located near Pavia (Italy) and consists of three thermophilic digesters (D1, D2 and D3) in series (Fig. 1) treating up to 120 kton/year of high-solid (about 18.5%) organic waste. Each digester has total and headspace volumes of approximately 4500 m^3 and less than 500 m^3, respectively. A direct injection of saturated steam (p = 4 bar) and insulation by a 10 cm polystyrene layer on the external wall maintain the digestate at 55 °C and the headspace at 45–50 °C in winter and 50–55 °C in summer. The feedstock is first collected in basins and then moved to a mixing unit to be heated and homogenized with the digestate from D3 and water resulting in a digestate total solid (TS) and volatile solid (VS) concentrations of 10.7% and 6.6%, respectively. A side-stream ammonia stripping unit is coupled to D1 to control ammonium concentration in the digestate (Fig. 1). At the end of the digestion process, the digestate is collected in a 50000 m^3 storage tank prior to being used as fertilizer. The biogas produced by the plant is converted by a combined heat and power (CHP) system to electrical energy and primarily consumed on-site (about 30%), while the excess is sent to the national grid. Biogas treatments, placed upstream the CHP system, include desulfurization, chilling, condensate removal, compression through blowers and purification with activated carbon filters.

2.2. Operational conditions

During the 81 weeks of the study (from January 2nd, 2017 to July 16th, 2018), the plant treated about 108 kton of feedstock under various organic loading rate (OLR) and hydraulic retention time (HRT) conditions (Table 1). The average composition of the feedstock included 90% of sewage sludge from municipal and industrial (agrifood industry) wastewater treatment plants and 10% of co-products (digestate and liquid fraction) of source segregated domestic food waste. H_2S levels ranging between 1200 and 2500 ppm were measured under completely anaerobic conditions in the biogas produced by a pilot-scale thermophilic digester fed with the same substrate and operated at OLRs and HRTs similar to those described in Table 1.

Prior to starting microaeration, the three full-scale digesters were operated under strict anaerobic conditions (RO < 0.1%) for six months. Air dosing (kNm^3/d) was adjusted manually to maintain the RO concentrations in the biogas at fixed values in the range of 0.2–2.0% (v/v) during six different operational periods (P1–P6, Table 1).

2.3. Monitoring and analytical methods

Online biogas production was measured by a ST51 mass flow meter (FCI, USA) installed as shown in Fig. 1. Biogas composition (CH_4, CO_2, H_2S and O_2) in the three digesters and upstream of the CHP unit was monitored six times/day using an Optima 7 portable biogas analyzer (MRU GmbH, Germany). Oxygen dosing to the headspace was performed by using type CL 10/01 (in D1) and CL 4/01 (in D2 and D3) side channel blowers (Mapro, Italy) and controlled by glass tube flowmeters (ASA, Italy). Feedstock and digestate samples were periodically collected for VS and TS analyses. Total ammonia nitrogen (TAN) and carbonate alkalinity (defined as the proton accepting capacity of the carbonate weak acid subsystem and indicated as H_2CO_3* alkalinity) concentrations in the digestate were measured after filtration through GF3 grade 1.2-μm glass microfiber filters (CHMLAB, Spain). TAN, TS and VS were measured according to standard methods (APHA, 2005). H_2CO_3* alkalinity was measured as described by Lahav et al. (2002).

3. Results and discussion

3.1. Desulfurization performance of the thermophilic digesters

The temporal profiles of RO and H_2S concentrations in the biogas produced by the three thermophilic digesters monitored in this study are shown in Fig. 2. The desulfurization efficiency was nearly 100% at all tested RO concentrations as the effluent H_2S levels in the biogas remained below 10 ppm (Table 2). In a previous study, Wu et al. (2016) studied microaerobic biogas desulfurization at mesophilic (35 °C) and thermophilic (55 °C) temperatures in a lab-scale fermenter treating rice straw. O_2 was supplied to the fermenter headspace at feed concentrations ranging from 0.5 to 10 times the stoichiometric requirement for partial H_2S oxidation to S^0. O_2 supply at a 4-fold higher concentration than the stoichiometric requirement resulted in RO values in biogas above 0.5% and desulfurization efficiencies of 93.8% and 94.5% under mesophilic and thermophilic conditions, respectively. O_2 supply below the stoichiometric requirement resulted in RO < 0.25% in biogas and desulfurization efficiencies below 75%. Similarly, Toth et al. (2015) observed a significant loss of H_2S removal efficiency (25–30%) in a

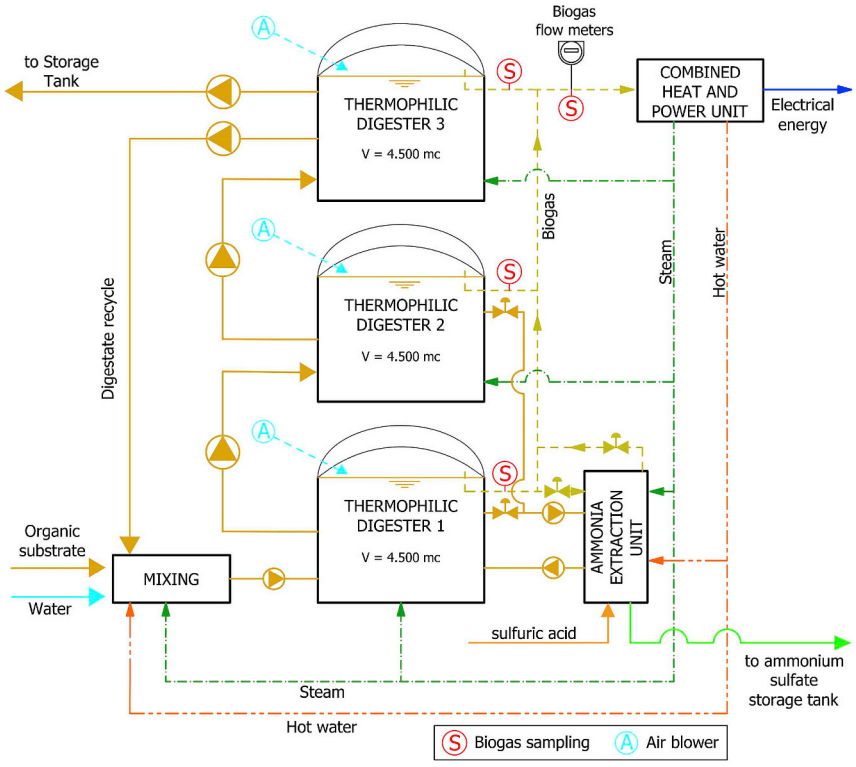


Fig. 1. Flow scheme of the full-scale thermophilic digestion plant.

Table 1
Microaerobic operation of the full-scale thermophilic digesters during the study.

Period	P1	P2	P3	P4	P5	P6
Operational time (weeks)	0–16	17–29	30–58	59–66	67–74	75–81
HRT (d)	71 ± 16	44 ± 4	58 ± 11	59 ± 5	56 ± 7	59 ± 7
Organic Loading Rate (OLR, kg VS/m^3 d)	1.5 ± 0.4	2.3 ± 0.3	1.9 ± 0.4	1.7 ± 0.1	1.8 ± 0.1	1.8 ± 0.1
Air flow (kNm^3/d)	0.71 ± 0.16	0.77 ± 0.13	0.56 ± 0.08	0.33 ± 0.03	0.19 ± 0.04	0.09 ± 0.02
Residual oxygen (RO, %)	2.0	1.3 ± 0.3	1.0	0.7	0.4 ± 0.1	0.2

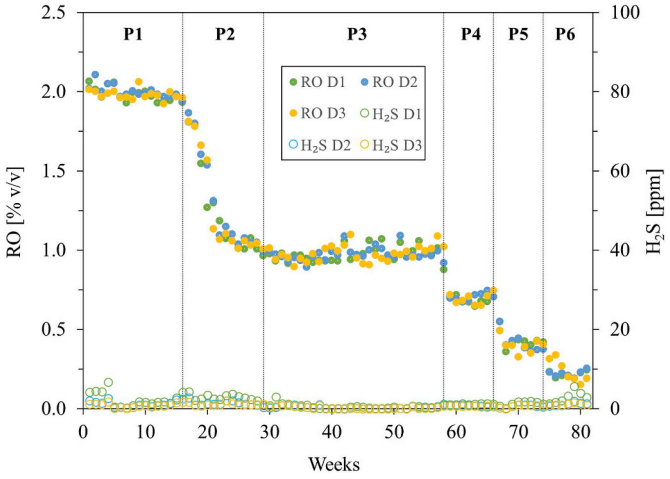


Fig. 2. Temporal profiles of residual oxygen (RO) and H_2S in the biogas of the three full-scale thermophilic digesters (D1, D2 and D3) during the study.

biotrickling filter treating sulfide-contaminated gas streams when the RO levels decreased below 1%.

In this study, high-efficiency desulfurization was maintained at RO concentrations in biogas as low as 0.2% and under OLRs and HRTs ranging from 1.5 to 2.3 kg VS/m^3 d and from 44 to 71 d, respectively (Table 1). Biogas production ranged between 8.5 and 12.3 ton/d (Table 2). Nevertheless, linear regressions were obtained between the RO values maintained in the headspace and both the air-to-biogas ratio (R^2 = 0.94) and specific oxygen supply (R^2 = 0.91) (Fig. 3). The intercepts of the two linear regressions on the y axis were 0.73 %ton air/ton biogas (standard error, SE = 0.17) and 0.88 kg O_2/ton VS_{in} (SE = 0.21) and provide an estimation of the oxygen consumption for the complete H_2S removal in the three digesters per ton of produced biogas and feed VS, respectively. Slopes were 5.02 (SE = 0.14) and 5.11 (SE = 0.18) for the air-to-biogas ratio and specific oxygen supply regressions, respectively. These values provide a valid support for designing the aeration systems of full-scale thermophilic digesters co-digesting high-solid sewage sludge and optimize oxygen dosing for desulfurization in order to reduce the aeration costs and improve biogas quality.

Diaz et al. (2010) supplemented 0.94 and 1.27 Nm^3 air/m^3 feed to the headspace of a 200 L mesophilic digester fed with sewage sludge, corresponding to a specific oxygen supply of approximately 7.9 and 11.9 g O_2/kg VS_{in}, respectively. The authors reported that only 30–40% of the total oxygen supplied to the system served for H_2S oxidation, while the rest left the bioreactor with the produced biogas. This means that only 2.3–3.5 g O_2/kg VS_{in} were used for H_2S removal from biogas. These values are 2.6–4.0 times higher than the specific oxygen consumption observed in our digesters (0.88 g O_2/kg VS_{in}) as the H_2S concentration reported by Diaz et al. (2010) in absence of microaeration (14 g H_2S/Nm^3 biogas corresponding to about 9300 ppm) was

Table 2
Biogas production and composition under the microaerobic conditions tested in this study.

Period	P1	P2	P3	P4	P5	P6
Biogas production (ton/d)	8.5 ± 2.0	12.3 ± 1.7	11.2 ± 1.9	10.2 ± 1.3	10.6 ± 0.6	9.1 ± 0.4
Total CH_4 production (kNm^3/d)	3.8 ± 0.9	6.1 ± 1.1	5.6 ± 1.1	4.9 ± 0.7	5.2 ± 0.3	4.5 ± 0.2
Specific CH_4 production (NL/kg VS_{in})	186 ± 12	194 ± 32	215 ± 17	218 ± 32	212 ± 12	186 ± 15
Methane (CH_4, %)	56.3 ± 2.1	60.4 ± 2.3	61.3 ± 1.6	61.6 ± 1.7	63.1 ± 2.6	63.4 ± 3.3
Carbon dioxide (CO_2, %)	34.0 ± 3.3	32.0 ± 2.6	33.3 ± 1.6	34.3 ± 1.8	33.6 ± 2.4	33.7 ± 3.3
Nitrogen (N_2, %) [a]	7.7 ± 1.1	6.4 ± 0.2	4.5	3.4 ± 0.1	3.0 ± 0.1	2.7
Residual oxygen (RO, %)	2.0	1.3 ± 0.3	1.0	0.7	0.4 ± 0.1	0.2
Hydrogen sulfide (H_2S, ppm)	< 10	< 10	< 10	< 10	< 10	< 10

[a] calculated as a complement to 100% based on CH_4, CO_2 and RO concentrations during each operational period.

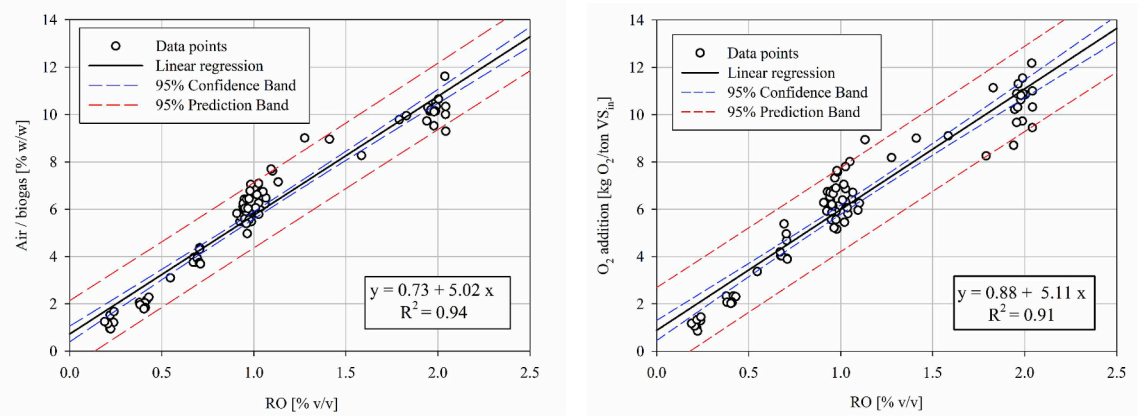


Fig. 3. Linear regressions (solid lines) of RO with air-to-biogas ratio (left) and specific oxygen supply (right) in the three full-scale thermophilic digesters. Dotted lines indicate 95% confidence (blue) and prediction (red) bands for each regression. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.7–7.8 fold higher than those measured during this study in the parallel pilot-scale thermophilic digester run under anaerobic conditions (1200–2500 ppm). The lower H_2S concentrations observed in this study can be attributed to the high ammonia concentrations commonly occurring during the AD of high-solid sewage sludge (Fricke et al., 2007).

Moestedt et al. (2013) showed that TAN and free ammonia nitrogen (FAN) levels above 1800 and 200 mg/L significantly reduce the abundance of SRB in the digester. In a recent study, Han et al. (2019) dosed NH_4Cl into mesophilic and thermophilic AD reactors as an inhibitor of H_2S production, resulting in a decrease of H_2S concentration in the biogas by over 50%. TAN concentrations in our digesters (3.5–3.8 g N–NH_4^+/kg digestate) were probably higher than those occurring in the mesophilic digester operated by Diaz et al. (2010) as this was run at a lower solid content (about 1.5% VS). FAN is extremely toxic to microorganisms and its fraction increases with pH and temperature (Westerholm et al., 2016). As a result, thermophilic operation results in higher FAN levels compared to mesophilic conditions and can considerably limit the SRB activity. In this study, the H_2S production by SRB and, thus, S^0 accumulation were presumably limited by the high TAN and FAN levels occurring in the thermophilic digesters. As a result, thermophilic operation can reduce the costs of digester maintenance and the frequency of cleaning operations that can disrupt the digestion process.

A total oxygen consumption of 1.49 L O_2/Nm^3 biogas was calculated based on the value of specific air consumption (0.73 %ton air/ton biogas) estimated in this study. According to Eq. (2), the oxidation of 1200–2500 ppm of H_2S to S^0 consumes 0.61–1.29 L O_2/Nm^3 biogas. This means that, assuming that all H_2S oxidized was converted to S^0 due to the limited O_2 availability in the headspace, the 13–59% of the consumed oxygen was not utilized for S^0 production. This is in agreement with Ramos et al. (2014a), reporting that between 43% and 55% of the O_2 supplied to the headspaces of two pilot-scale mesophilic digesters was used for processes other than H_2S oxidation to S^0. It is possible that the excess O_2 was consumed for S^0 oxidation to SO_4^{2-} and/or by facultative/microaerophilic bacteria degrading the organic substrate as described by Jenicek et al. (2014).

The deposition of S^0 onto gas pipes and internal walls of the digester causes clogging and reduces the biogas storage capacity of the headspace, resulting in a negative impact on desulfurization efficiency (Diaz and Fdz-Polanco, 2012). Ramos et al. (2014a) recommended a maximum cleaning interval of 14 months to maintain a high O_2 usage efficiency and reduce the cost of microaeration in two pilot-scale digesters with a produced H_2S concentration in the biogas of 5000 ppm. According to Jenicek et al. (2017), strict air dosing into the liquid phase reduces sulfur deposition and clogging events as less oxygen reaches the headspace. Although in this work microaeration took place in the headspace solely, no issues linked to S^0 accumulation occurred during the 81 operational weeks of the study. This can be linked to the limited H_2S production in the digesters due to high TAN levels as well as to excess oxygen consumption for S^0 oxidation to SO_4^{2-}.

3.2. Biogas production and composition under different microaerobic conditions

Microaeration showed no detrimental effects on thermophilic digestion at the different RO conditions tested. Due to the OLR and HRT fluctuations (Table 1), the total biogas and CH_4 production ranged from 8.5 to 12.3 ton/d and from 3.8 to 6.1 kNm^3/d, respectively (Table 2). The highest specific CH_4 production (218 NL/kg VS_{in}) was obtained at a RO concentration of 0.7% (P4), while a minimum production of 186 NL/kg VS_{in} was observed at RO concentrations of 2.0% (P1) and 0.2% (P6) (Table 2). CH_4 concentration in biogas increased from 56.3% in P1 to 63.4% in P6, whereas CO_2 concentration was relatively stable during the study. This resulted in a 13% increase of the CH_4/CO_2 ratio in the produced biogas between P1 (RO = 2.0%) and P6 (RO = 0.2%) (Table 2). The microaerobic conditions and high TAN levels in the three

digesters likely favored hydrogenotrophic over acetoclastic methanogenesis, as hydrogenotrophic methanogens possess a higher tolerance to oxygen (Tang et al., 2004) and ammonia (Angelidaki and Ahring, 1993), which provides them a competitive advantage in microaerobic and ammonia-rich systems.

In previous studies, microaeration was reported to reduce sulfide toxicity to methanogens (Jeníček et al., 2008) and enhance hydrolysis (Johansen and Bakke, 2006; Montalvo et al., 2016) and the aerobic biodegradation of the volatile fatty acids (VFA) produced during acidogenesis (Jeníček et al., 2014). However, microaeration reduces the CH_4 content of biogas and specific CH_4 production due to biogas dilution and aerobic biodegradation of organic matter, respectively (Jeníček et al., 2017). In this study, decreasing the RO concentration in biogas between 2.0% and 0.2% reduced dilution and increased the CH_4 content of biogas with no detrimental effects on the H_2S removal performance of the three thermophilic digesters (Table 2). CO_2 in the headspace is in equilibrium with bicarbonate alkalinity in the digestate and depends also on pH and temperature of the liquid phase (Parkin and Owen, 1986). During the study, CO_2 concentration in the biogas was not expected to vary significantly as both pH and H_2CO_3* alkalinity (which can be approximated to bicarbonate alkalinity at the measured pH values) in the digestate were stable at 7.86 ± 0.08 and 9.93 ± 0.91 g $CaCO_3$/L, respectively, as a result of a balanced digestion process.

A 17% increase in the specific CH_4 production was observed when the RO concentration in the digesters was reduced from 2.0% (P1) to 0.7% (P4). The further RO decrease to 0.2% (P6) reduced the specific CH_4 production to the initial value (186 NL/kg VS_{in}, Table 2). Variations in specific CH_4 production can be mainly attributed to fluctuations in the feedstock composition, which commonly occur during full-scale plant operation. Additionally, the aerobic substrate biodegradation may have also influenced CH_4 productivity. It is possible that during P1 (RO = 2.0%) more oxygen was available to sustain both hydrolysis and VFA degradation by facultative bacteria. Decreasing the RO values below 2.0% likely reduced the aerobic VFA biodegradation and increased CH_4 productivity, while RO values below 0.7% may have reduced the hydrolytic activity of aerobic bacteria resulting in a lower specific CH_4 production.

3.3. Practical implications and operational remarks

The results of this study show that RO can be reliably used as control parameter for high-efficiency H_2S removal from biogas in full-scale thermophilic digesters co-digesting sewage sludge. Maintaining RO concentrations ≥0.2% in biogas resulted in complete desulfurization under fluctuating OLR and HRT conditions (Table 1). Single measurements of RO concentration (as performed in this study) are preferable to online monitoring especially when H_2S levels in biogas are high in order to avoid sensor poisoning due to long-term exposure.

Biogas upgrading mainly consists in the removal of CO_2 and trace components including H_2S, O_2 and N_2 to meet the standards for biomethane production. The European standard EN 16726 sets the CO_2, O_2 and H_2S content in biomethane for pipeline utilization in the range of 2.5–4%, 0.001–1% and ≤5 ppm, respectively. The removal of O_2 and N_2 from biogas is expensive and usually performed using membranes or low-temperature pressure swing adsorption (PSA). As a result, dosing pure oxygen rather than air is preferable for biomethane production, as it reduces the post-treatment costs. Our results show that dosing 0.88 kg of pure oxygen per ton of VS_{in} in the three full-scale thermophilic digesters operated in this study would produce oxygen-free biogas and reduce the costs of biomethane production. Additionally, the specific electric consumption during the study was of only 3.4 kWh/ton biogas, lower than the value of 7.8 kWh/ton biogas calculated based on the data reported by Diaz et al. (2015) for biogas desulfurization in three full-scale digesters via microaeration. This confirms microaerobic desulfurization as an inexpensive solution for biogas upgrading.

Introducing air into the headspace of a digester can be considered safe as far as the RO concentration does not exceed the limiting oxygen concentration (LOC) of the produced biogas. Dupont and Accorsi (2006) studied the flammability of biogas mixtures composed of 50% CH_4 and 50% CO_2 and determined LOC values of 11.1% and 13.2% for vapor-saturated and dry biogas, respectively. These values are well above the RO concentrations (0.2–2.0%) maintained in the three digesters during this study. However, biogas leaks from the digester represent a concern and necessary safety measures, especially during maintenance operations (e.g. cleaning of digesters and biogas storage tank repair), should be implemented to minimize the risk of explosion.

4. Conclusions

- Complete biogas desulfurization was achieved in three full-scale thermophilic digesters in series over 81 weeks under various RO (0.2–2.0%), OLR (1.5–2.3 kg VS/m^3 d^1) and HRT (44–71 d) conditions.
- The linear relationship between the O_2 addition in the headspace and RO concentration in biogas allowed to estimate a specific oxygen consumption of 0.88 kg O_2/ton VS_{in} (corresponding to 1.49 L O_2/Nm^3 biogas) for the complete H_2S removal from biogas.
- From 13% to 59% of the O_2 consumed during the process was not utilized for H_2S conversion to S^0.
- The low specific oxygen consumption for H_2S removal observed in this study (1.49 L O_2/Nm^3 biogas) was presumably due to high TAN (3.5–3.8 g N–NH_4^+/kg digestate) and FAN concentrations in digestate limiting H_2S production by SRB.
- No significant S^0 accumulation in the headspaces of the three digesters was observed during the study.
- CO_2 concentration in the biogas remained stable (32.0–34.3%) at the different microaerobic conditions investigated, being CO_2 in equilibrium with the pH and bicarbonate alkalinity of the digestate, which did not vary significantly due to a balanced digestion process.

Acknowledgements

This work was supported by Programma di Sviluppo Rurale (PSR) Campania 2014/2020 - METAGRO Project and Programma Operativo Nazionale (PON) 2014/2020 – BIOFEEDSTOCK Project. The Authors would like to thank Ms. Sara Panico and Mr. Alfonso Bova for their support during the whole experimental campaign.

References

Abatzoglou, N., Boivin, S., 2009. A review of biogas purification processes. Biofuels, Bioprod. Biorefining 3, 42–71. https://doi.org/10.1002/bbb.117.

Angelidaki, I., Ahring, B.K., 1993. Thermophilic anaerobic digestion of livestock waste: the effect of ammonia. Appl. Microbiol. Biotechnol. 38, 560–564. https://doi.org/10.1007/BF00242955.

APHA, 2005. Standard Methods for the Examination of Water and Wastewater. American Water Works Association/American Public Works Association/Water Environment Federation.

Awe, O.W., Zhao, Y., Nzihou, A., Minh, D.P., Lyczko, N., 2017. A review of biogas utilisation, purification and upgrading technologies. Waste Biomass Valorization 8, 267–283. https://doi.org/10.1007/s12649-016-9826-4.

Di Capua, F., Ahoranta, S.H., Papirio, S., Lens, P.N.L., Esposito, G., 2016. Impacts of sulfur source and temperature on sulfur-driven denitrification by pure and mixed cultures of *Thiobacillus*. Process Biochem. 51, 1576–1584. https://doi.org/10.1016/j.procbio.2016.06.010.

Di Capua, F., Pirozzi, F., Lens, P.N.L., Esposito, G., 2019. Electron donors for autotrophic denitrification. Chem. Eng. J. 362, 922–937. https://doi.org/10.1016/J.CEJ.2019.01.069.

Diaz, I., Fdz-Polanco, M., 2012. Robustness of the microaerobic removal of hydrogen sulfide from biogas. Water Sci. Technol. 65, 1368–1374. https://doi.org/10.2166/wst.2012.013.

Diaz, I., Lopes, A.C., Perez, S.I., Fdz-Polanco, M., 2010. Performance evaluation of oxygen, air and nitrate for the microaerobic removal of hydrogen sulphide in biogas from sludge digestion. Bioresour. Technol. 101, 7724–7730. https://doi.org/10.1016/j.biortech.2010.04.062.

Diaz, I., Perez, S.I., Ferrero, E.M., Fdz-Polanco, M., 2011. Effect of oxygen dosing point and mixing on the microaerobic removal of hydrogen sulphide in sludge digesters.

Bioresour. Technol. 102, 3768–3775. https://doi.org/10.1016/j.biortech.2010.12.016.
Diaz, I., Ramos, I., Fdz-Polanco, M., 2015. Economic analysis of microaerobic removal of H_2S from biogas in full-scale sludge digesters. Bioresour. Technol. 192, 280–286. https://doi.org/10.1016/j.biortech.2015.05.048.
Dupont, L., Accorsi, A., 2006. Explosion characteristics of synthesised biogas at various temperatures. J. Hazard Mater. B136, 520–525. https://doi.org/10.1016/j.jhazmat.2005.11.105.
Fricke, K., Santen, H., Wallmann, R., Hüttner, A., Dichtl, N., 2007. Operating problems in anaerobic digestion plants resulting from nitrogen in MSW. Waste Manag. 27, 30–43. https://doi.org/10.1016/J.WASMAN.2006.03.003.
Guerrero, L., Montalvo, S., Huiliñir, C., Campos, J.L., Barahona, A., Borja, R., 2015. Advances in the biological removal of sulphides from aqueous phase in anaerobic processes : a review. Environ. Rev. 24, 84–100. https://doi.org/10.1139/er-2015-0046.
Hall, J.R., Schoenfisch, M.H., 2018. Direct electrochemical sensing of hydrogen sulfide without sulfur poisoning. Anal. Chem. 90, 5194–5200. https://doi.org/10.1021/acs.analchem.7b05421.
Han, Y., Qu, Q., Li, J., Zhuo, Y., Zhong, C., Peng, D., 2019. Performance of ammonium chloride dosage on hydrogen sulfide in-situ prevention during waste activated sludge anaerobic digestion. Bioresour. Technol 276, 91–96. https://doi.org/10.1016/j.biortech.2018.12.089.
Jenicek, P., Celis, C.A., Krayzelova, L., Anferova, N., Pokorna, D., 2014. Improving products of anaerobic sludge digestion by microaeration. Water Sci. Technol. 69, 803–809. https://doi.org/10.2166/wst.2013.779.
Jenicek, P., Horejs, J., Pokorna-Krayzelova, L., Bindzar, J., Bartacek, J., 2017. Simple biogas desulfurization by microaeration – full scale experience. Anaerobe 46, 41–45. https://doi.org/10.1016/j.anaerobe.2017.01.002.
Jenicek, P., Keclik, F., Maca, J., Bindzar, J., 2008. Use of microaerobic conditions for the improvement of anaerobic digestion of solid wastes. Water Sci. Technol. 58, 1491–1496. https://doi.org/10.2166/wst.2008.493.
Johansen, J.-E., Bakke, R., 2006. Enhancing hydrolysis with microaeration. Water Sci. Technol. 53, 43–50. https://doi.org/10.2166/wst.2006.234.
Khanal, S.K., Shang, C., Huang, J.C., 2003. Use of ORP (oxidation-reduction potential) to control oxygen dosing for online sulfide oxidation in anaerobic treatment of high sulfate wastewater. Water Sci. Technol. 47, 183–189. https://doi.org/10.2166/wst.2003.0645.
Khanongnuch, R., Di Capua, F., Lakaniemi, A.-M., Rene, E.R., Lens, P.N.L., 2018. H_2S removal and microbial community composition in an anoxic biotrickling filter under autotrophic and mixotrophic conditions. J. Hazard Mater. 367, 397–406. https://doi.org/10.1016/J.JHAZMAT.2018.12.062.
Kobayashi, T., Li, Y.Y., Kubota, K., Harada, H., Maeda, T., Yu, H.Q., 2012. Characterization of sulfide-oxidizing microbial mats developed inside a full-scale anaerobic digester employing biological desulfurization. Appl. Microbiol. Biotechnol. 93, 847–857. https://doi.org/10.1007/s00253-011-3445-6.
Krayzelova, L., Bartacek, J., Diaz, I., Jeison, D., Volcke, E.I.P., Jenicek, P., 2015. Microaeration for hydrogen sulfide removal during anaerobic treatment: a review. Rev. Environ. Sci. Biotechnol. 14, 703–725. https://doi.org/10.1007/s11157-015-9386-2.
Krayzelova, L., Bartacek, J., Kolesarova, N., Jenicek, P., 2014. Microaeration for hydrogen sulfide removal in UASB reactor. Bioresour. Technol. 172, 297–302. https://doi.org/10.1016/j.biortech.2014.09.056.
Lahav, O., Morgan, B., Loewenthal, R.E., 2002. Rapid, simple, and accurate method for measurement of VFA and carbonate alkalinity in anaerobic reactors. Environ. Sci. Technol. 36, 2736–2741. https://doi.org/10.1021/es011169v.
Mancino, G., Cimino, S., Lisi, L., 2016. Sulphur poisoning of alumina supported Rh catalyst during dry reforming of methane. Catal. Today 277, 126–132. https://doi.org/10.1016/J.CATTOD.2015.10.035.
Moestedt, J., Nilsson Paledal, S., Schnurer, A., 2013. The effect of substrate and operational parameters on the abundance of sulphate-reducing bacteria in industrial anaerobic biogas digesters. Bioresour. Technol. 132, 327–332. https://doi.org/10.1016/j.biortech.2013.01.043.
Montalvo, S., Huilinir, C., Ojeda, F., Castillo, A., Lillo, L., Guerrero, L., 2016. Microaerobic pretreatment of sewage sludge: effect of air flow rate, pretreatment time and temperature on the aerobic process and methane generation. Int. Biodeterior. Biodegrad. 110, 1–7. https://doi.org/10.1016/J.IBIOD.2016.01.010.
Nghiem, L.D., Manassa, P., Dawson, M., Fitzgerald, S.K., 2014. Oxidation reduction potential as a parameter to regulate micro-oxygen injection into anaerobic digester for reducing hydrogen sulphide concentration in biogas. Bioresour. Technol. 173, 443–447. https://doi.org/10.1016/j.biortech.2014.09.052.
Ortloff, F., Bohnau, J., Kramar, U., Graf, F., Kolb, T., 2016. Studies on the influence of H_2S and SO_2 on the activity of a PdO/Al_2O_3 catalyst for removal of oxygen by total oxidation of (bio-)methane at very low O_2:CH_4 ratios. Appl. Catal. B Environ. 182, 550–561. https://doi.org/10.1016/J.APCATB.2015.09.026.
Oude Elferink, S.J.W.H., Visser, A., Hulshoff Pol, L.W., Stams, A.J.M., 1994. Sulfate reduction in methanogenic bioreactors. FEMS Microbiol. Rev. 15, 119–136. https://doi.org/10.1111/j.1574-6976.1994.tb00130.x.
Pandey, S.K., Kim, K.-H., Tang, K.-T., 2012. A review of sensor-based methods for monitoring hydrogen sulfide. TrAC Trends Anal. Chem. 32, 87–99. https://doi.org/10.1016/J.TRAC.2011.08.008.
Papirio, S., Esposito, G., Pirozzi, F., 2013a. Biological inverse fluidized-bed reactors for the treatment of low pH- and sulphate-containing wastewaters under different COD conditions. Environ. Technol. 34, 1141–1149. https://doi.org/10.1080/09593330.2012.737864.
Papirio, S., Villa-Gomez, D.K., Esposito, G., Pirozzi, F., Lens, P.N.L., 2013b. Acid mine drainage treatment in fluidized-bed bioreactors by sulfate-reducing bacteria: a critical review. Crit. Rev. Environ. Sci. Technol. 43, 2545–2580. https://doi.org/10.1080/10643389.2012.694328.
Parkin, G.F., Owen, W.F., 1986. Fundamentals of anaerobic digestion of wastewater sludges. J. Environ. Eng. 112, 867–920. https://doi.org/10.1061/(asce)0733-9372(1986)112:5(867).
Ramos, I., Fdz-Polanco, M., 2014. Microaerobic control of biogas sulphide content during sewage sludge digestion by using biogas production and hydrogen sulphide concentration. Chem. Eng. J. 250, 303–311. https://doi.org/10.1016/j.cej.2014.04.027.
Ramos, I., Perez, R., Fdz-Polanco, M., 2014a. The headspace of microaerobic reactors: sulphide-oxidising population and the impact of cleaning on the efficiency of biogas desulphurisation. Bioresour. Technol. 158, 63–76. https://doi.org/10.1016/j.biortech.2014.02.001.
Ramos, I., Perez, R., Reinoso, M., Torio, R., Fdz-Polanco, M., 2014b. Microaerobic digestion of sewage sludge on an industrial-pilot scale: the efficiency of biogas desulphurisation under different configurations and the impact of O_2 on the microbial communities. Bioresour. Technol. 164, 338–346. https://doi.org/10.1016/j.biortech.2014.04.109.
Rasi, S., Lantela, J., Rintala, J., 2011. Trace compounds affecting biogas energy utilisation - a review. Energy Convers. Manag. 52, 3369–3375. https://doi.org/10.1016/j.enconman.2011.07.005.
Stams, A.J.M., Plugge, C.M., de Bok, F.A.M., van Houten, B.H.G.W., Lens, P., Dijkman, H., Weijma, J., 2005. Metabolic interactions in methanogenic and sulfate-reducing bioreactors. Water Sci. Technol. 52, 13–20. https://doi.org/10.2166/wst.2005.0493.
Tang, Y., Shigematsu, T., Ikbal Morimura, S., Kida, K., 2004. The effects of micro-aeration on the phylogenetic diversity of microorganisms in a thermophilic anaerobic municipal solid-waste digester. Water Res. 38, 2537–2550. https://doi.org/10.1016/J.WATRES.2004.03.012.
Tilahun, E., Sahinkaya, E., Calli, B., 2018. A hybrid membrane gas absorption and bio-oxidation process for the removal of hydrogen sulfide from biogas. Int. Biodeterior. Biodegrad. 127, 69–76. https://doi.org/10.1016/J.IBIOD.2017.11.015.
Toth, G., Nemestothy, N., Belafi-Bako, K., Vozik, D., Bakonyi, P., 2015. Degradation of hydrogen sulfide by immobilized *Thiobacillus thioparus* in continuous biotrickling reactor fed with synthetic gas mixture. Int. Biodeterior. Biodegrad. 105, 185–191. https://doi.org/10.1016/J.IBIOD.2015.09.006.
Westerholm, M., Moestedt, J., Schnürer, A., 2016. Biogas production through syntrophic acetate oxidation and deliberate operating strategies for improved digester performance. Appl. Energy 179, 124–135. https://doi.org/10.1016/j.apenergy.2016.06.061.
Wu, M., Zhang, Y., Ye, Y., Lin, C., 2016. In situ removal of hydrogen sulfide during biogas fermentation at microaerobic condition. Appl. Biochem. Biotechnol. 180, 817–825. https://doi.org/10.1007/s12010-016-2135-3.